

Recursive Bootstrapping Data Collapse

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1 Introduction

Bootstrapping is a resampling technique where a dataset is repeatedly sampled with replacement from the original dataset to create multiple bootstrapped samples. It works as variances computed under the bootstrap, or empirical distribution, are good approximations of variances computed under the true data. This method is used when the theoretical distribution of an estimator is complex or unknown. The basic bootstrap involves resampling, computing replicates, and estimating the sampling distribution. Uses of the bootstrap are estimation of standard error, construction of confidence intervals, normal approximation intervals, and percentile intervals. Bootstrap methods have the asymptotic properties of consistency, since the bootstrap distribution converges to the true sampling distribution, and second order accuracy as bootstrap confidence intervals often have higher accuracy in comparison to traditional methods especially for smaller samples.

The recursive bootstrap is a bootstrapping technique where resampling with replacement is applied repeatedly with a recursive framework. This applies the technique multiple times, which can increase the stability or accuracy of a model.

This form of data manipulation is used by various artificial intelligence algorithms. The recursive bootstrap has been used in structure learning of Bayesian networks. The bootstrap method is used to estimate uncertainty in the structure; however, the recursive bootstrap goes further by repeatedly resampling the data. This stabilizes structure learning. The recursive bootstrap is being introduced to artificial general intelligence models to improve recursive learning abilities and long-term stability in neural networks. A form of Fractal Flux Bootstrap Time Spiral was introduced as a form of neural network that evolves recursively with bootstrap causality (however, this model uses fractal complexity to prevent model collapse).

2 Variables

This collapse is measured through changes in mean, standard deviation, variance, z-score, and skewness to measure the changes in data after using a resampling with replacement bootstrapping technique.

Changes in mean demonstrate a shift in the overall behavior of the data. This alters the average value as a result of the shifting of the overall shape of the data. The loss of outlier data shifts the mean towards a median value, increasing for data with a right skew and decreasing for data that is left skewed.

Changes in standard deviation and variance demonstrate the collapse of the function towards a single-point distribution. The standard deviation measures how far the data is spread from the mean. The variance is the square of standard deviation, which measures how much the data points differ from the mean. The decay of these variables demonstrates the collapse of unique data after the use of resampling with replacement bootstrapping.

Changes in the Z-score represent changes in the spread of the data. A small Z-score represents varied data spread far from the mean, which is unpredictable. A large Z-score represents predictable data that is close to the mean and lacks variation.

Changes in skewness demonstrate the shifting of the behavior of the data. Instead of retaining the shape of the original dataset, the nature of resampling and replacement drifts towards a normal distribution.

3 Changes in Distribution

To test the limitations in resampling and replacement associated with bootstrapping, a synthetic data set with a right skewed distribution was generated, shown in Figure 1.

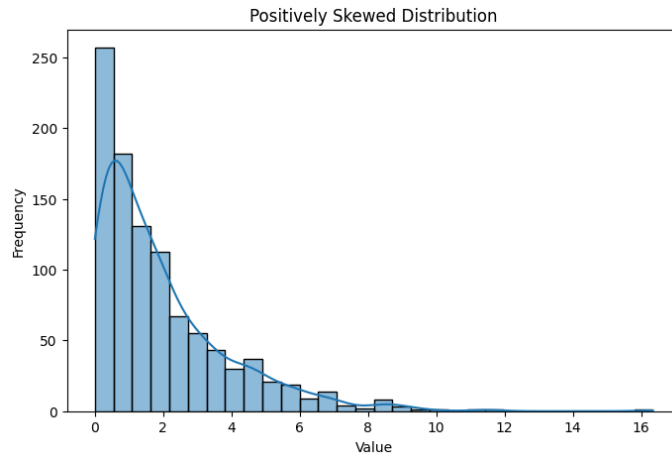


Figure 1: Original Synthetic Right Skewed Dataset

After multiple iterations of the bootstrap, as a result of resampling with replacement, the data will start losing its initial right skew and converge to a single point distribution. This follows the central limit theorem, which states that as the sample size increases, data begin to follow a normal shape. So, as the number of times the data is resampled increases, the data will begin to converge to a normal distribution. This does not occur due to sample size change, as bootstrapping generates data of identical sizes, but as a result of a larger number of iterations. Figure 2 demonstrates this change among the data.

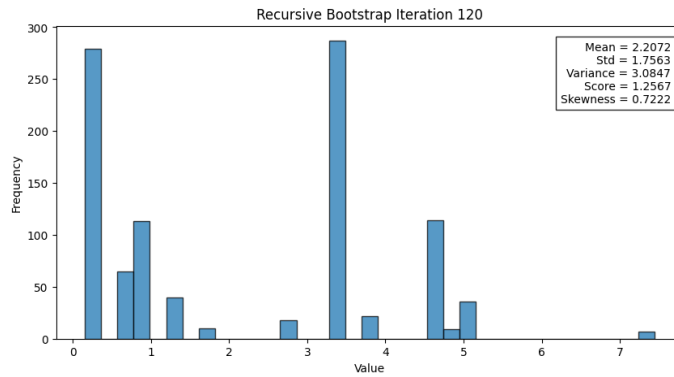


Figure 2: Dataset after 120 iterations of the Bootstrap

If the resampling technique is over-used, the data will begin to conform to a single-point distribution as a result of a loss in variance. This is demonstrated by Figure 3, which illustrates the initial right-skewed distribution collapsing to repeated points. This occurs as every iteration of the bootstrap causes data to be lost as a result of resampling and replacement techniques.

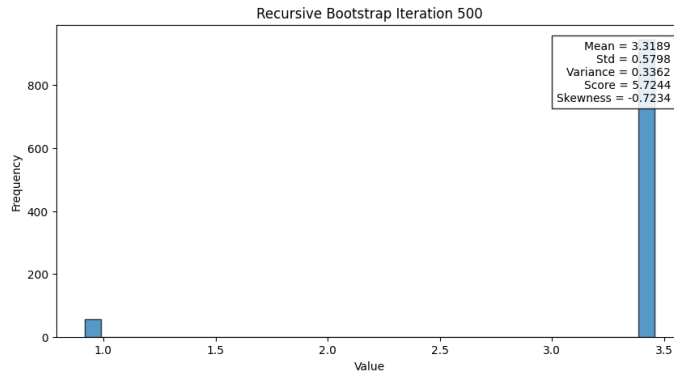


Figure 3: Dataset after 500 iterations of the Bootstrap

4 Changes in Statistics

Over recursive bootstrap iterations, the mean, standard deviation, variance, z-score, and skewness of a dataset will change as a result of the alteration of the data. Since outliers in the dataset continue to be removed due to resampling with replacement techniques, the spread of the data will decrease, impacting a variety of measurements on the data.

Figure 4 illustrates the change that occurs in the mean of the distribution over iterations of the bootstrap.



Figure 4: Change in mean over recursive bootstrap iterations

Figure 5 illustrates the change that occurs in the standard deviation of the distribution over iterations of the bootstrap.

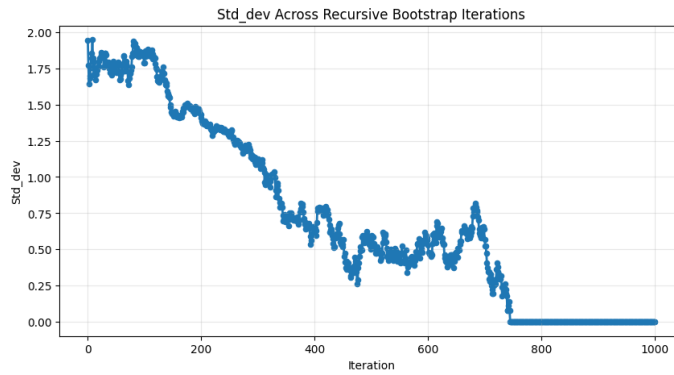


Figure 5: Change in standard deviation over recursive bootstrap iterations

Figure 6 illustrates the change that occurs in the variance of the distribution over iterations of the bootstrap.

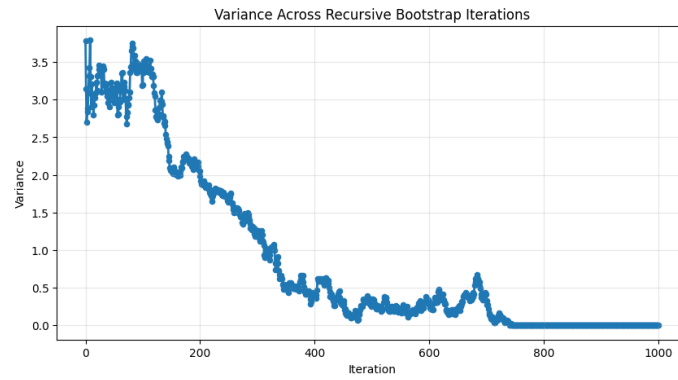


Figure 6: Change in variance over recursive bootstrap iterations

Figure 7 illustrates the change that occurs in the z-score of the distribution over iterations of the bootstrap.

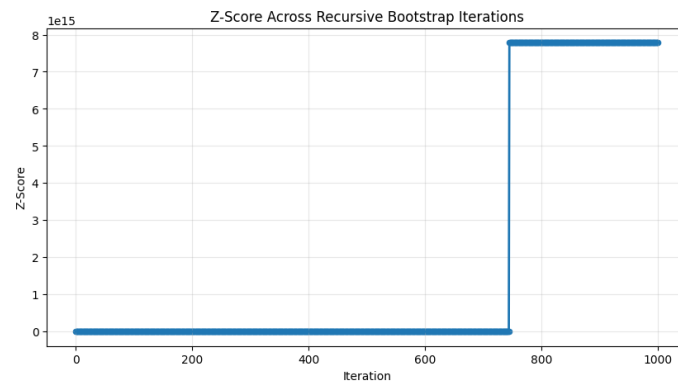


Figure 7: Change in Z-Score over recursive bootstrap iterations

Figure 8 illustrates the change that occurs in the skewness of the distribution over iterations of the bootstrap.



Figure 8: Change in Skewness over recursive bootstrap iterations

5 Modeling Changes in Variance, Standard Deviation, and Mean

The variance collapse model is shown below.

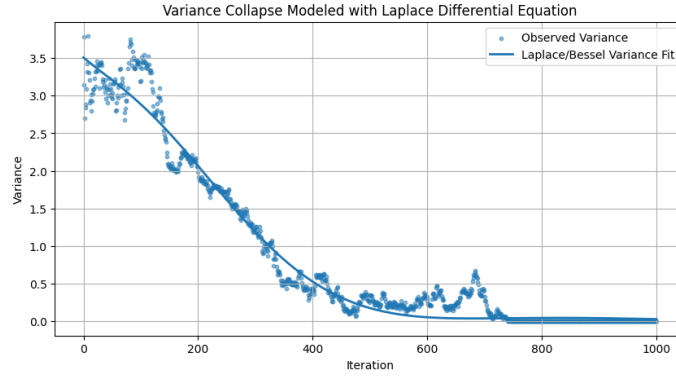


Figure 9: Graph of fitted Variance Collapse Differential Equation Model from (1)

The standard deviation collapse model is shown below.

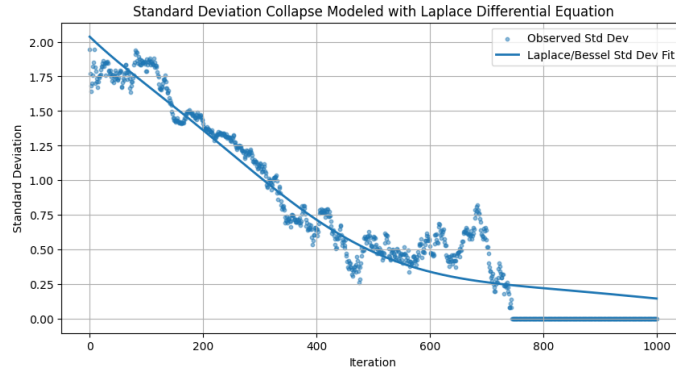


Figure 10: Graph of fitted Standard Deviation Collapse Differential Equation Model from (2)

The mean change model is shown below.

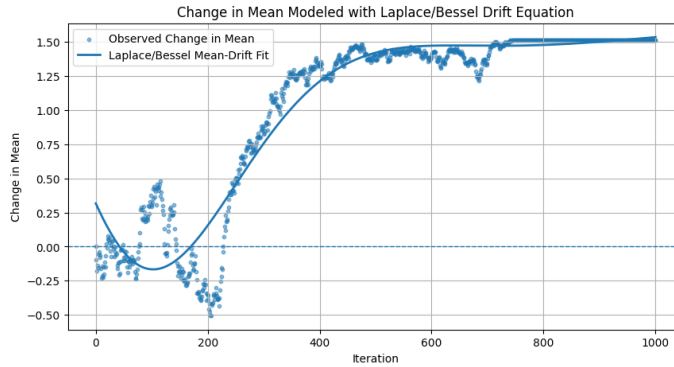


Figure 11: Graph of fitted Change in Mean Differential Equation Model from (3)

The second order spring mass differential equation accounts for oscillatory behavior in exponential collapse. This can resemble restoring forces and instability that occur from a point being increasingly resampled, decreasingly resampled, or not resampled at all.

6 Significance

Recursive bootstrapping destroys information over time, leading to decay in quality of returned data. Thus, models and neural networks that use recursive bootstrapping can iteratively cause datasets to lose diversity. So when creating a neural network based off recursive bootstrap techniques it must be used in combination with something that can reintroduce variance to the system. Along with this, algorithms based in the recursive bootstrap will change properties of the initial dataset.

7 References

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